- **Rutherford (Rd)** is defined as the quantity of a radioactive substance which produces 10^6 disintegrations in one second.
  1 Rd = 10^6 disintegrations per second.
- The SI unit of radioactivity is **becquerel.** It is defined as the quantity of one disintegration per second.
- Helium nucleus ($_2He^4$) consisting of two protons and two neutrons is known as alpha particle.
- Beta particles are electrons ($_{-1}e^0$), which are the basic elementary particles present in all atoms.
- Gamma rays are electromagnetic waves consisting of photons.
- A nuclear reaction in which an unstable parent nucleus emits an alpha particle and forms a stable daughter nucleus is called as 'alpha decay'.
- A nuclear reaction in which an unstable parent nucleus emits a beta particle and forms a stable daughter nucleus is called as 'beta decay'.
- The process of breaking (splitting) up of a heavier nucleus into two smaller nuclei with the release of a large amount of energy is called **'nuclear fission'.**
- The energy released in a nuclear fission process is about 200 MeV.
- There are some radioactive elements which can be converted into a fissionable material. They are called as '**fertile materials**'. e.g. Uranium-238, Thorium-232, Plutonium-240.
- Controlled chain reaction is used in a nuclear reactor to produce energy in a sustained and controlled manner.
- The process in which two lighter nuclei combine to form a heavier nucleus is termed as **'nuclear fusion'.**
- Nuclear fusion or thermonuclear reaction is the source of light and heat energy in the Sun and other stars.
- The safe limit of receiving the radiation is about 100 mR per week.

TEXTBOOK EVALUATION

I. Choose the correct answer

1. Man-made radioactivity is also known as _____
   a. Induced radioactivity
   b. Spontaneous radioactivity
   c. Artificial radioactivity
   d. a & c
2. Unit of radioactivity is _____
   a. roentgen b. curie
   c. becquerel d. all the above
3. Artificial radioactivity was discovered by _____
   a. Bequerel b. Irene Curie
   c. Roentgen d. Neils Bohr
4. In which of the following, no change in mass number of the daughter nuclei takes place
   i) α decay ii) β decay
   iii) γ decay iv) neutron decay
   a. (i) is correct
   b. (ii) and (iii) are correct
   c. (i) & (iv) are correct
   d. (ii) & (iv) are correct
5. _____ isotope is used for the treatment of cancer.

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